

Osaid Khan

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Dear Hiring Team,

It is with great pleasure that I submit my candidacy for the **Software Engineer, Content Engineering (AI Content Orchestration)** role with **Paramount**. I bring over three years of professional experience in backend engineering, distributed systems, and AI orchestration, along with a strong academic foundation as a Master of Science in Computer Science graduate from Pace University, NY.

Through my experience across engineering roles, I have built and delivered production-grade backend systems that directly reflect **the complexities of building scalable, asynchronous orchestration services that power an “AI Editor”**. As a Software Engineer at Sperse, I architected a concurrent multi-agent orchestration system using LangChain and LangGraph, enabling scalable asynchronous workers and coordination across event-driven workflows — requiring rigorous system design to manage agentic action lifecycles. To ensure high performance in production, I engineered low-latency inference paths by optimizing prompt routing and agent concurrency, reducing end-to-end latency by 60% and lowering API costs by 50%. Prior to Sperse, as a Software Engineer at Qualitest, I contributed to the backbone infrastructure that powers critical data operations. I designed and developed a cloud-agnostic Storage Service using Python, PostgreSQL, and Redis that seamlessly managed high-volume metadata and payload delivery across GCP and AWS S3 — giving me direct, hands-on experience with the multi-cloud backend architectures and data contracts essential for video infrastructure. I also developed backend workflows and asynchronous task queues using AWS Step Functions and Lambda to handle long-running data processing jobs, which required **the kind of systems-level thinking and operational stability focus this role demands**. Together, these roles have prepared me to **design and implement backend solutions that manage the lifecycle of high-impact AI-generated clips** and drive impact within Paramount’s Content Engineering team.

My academic research experience has given me a rigorous foundation in model development, GPU-accelerated computing, and technical communication — skills that translate directly into **Paramount’s commitment to Excellence and Integrity in storytelling**. As an AI Researcher at the Pace Artificial Intelligence Lab, I conducted applied research on StyleGAN and Stable Diffusion for automated design generation, leveraging CUDA, PyTorch, and LoRA on an HPC cluster for high-volume model training. I was responsible for integrating ML models into production pipelines using vLLM for optimized inference and Hugging Face for scalable API-based delivery — ensuring both low-latency performance and reliable data delivery. Beyond building the systems themselves, I have experience communicating complex technical concepts, which aligns with **the cross-functional collaboration required to bridge ML workflows with video operations and encoding pipelines**. This experience has shaped me into an engineer who builds with rigor and communicates with clarity, both of which I would bring to **Paramount’s Content Engineering team and its mission to scale content intelligence**.

I have always been passionate about **unleashing the power of content and pursuing excellence through innovation** and envision myself contributing my talent to the furthering of **Paramount’s goal of entertaining the planet through exceptional storytelling**. I would love to discuss my interest in this position with your team and can be reached at +1 (945) 304-5781 or khanosaid726@gmail.com. Thank you for your time and consideration, I look forward to connecting!

Sincerely,
Osaid Khan